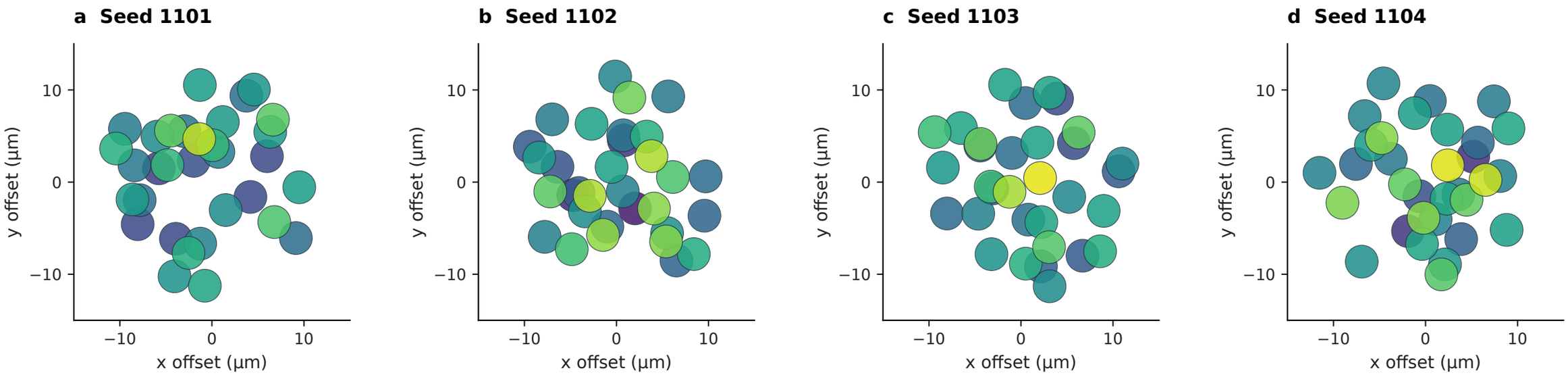
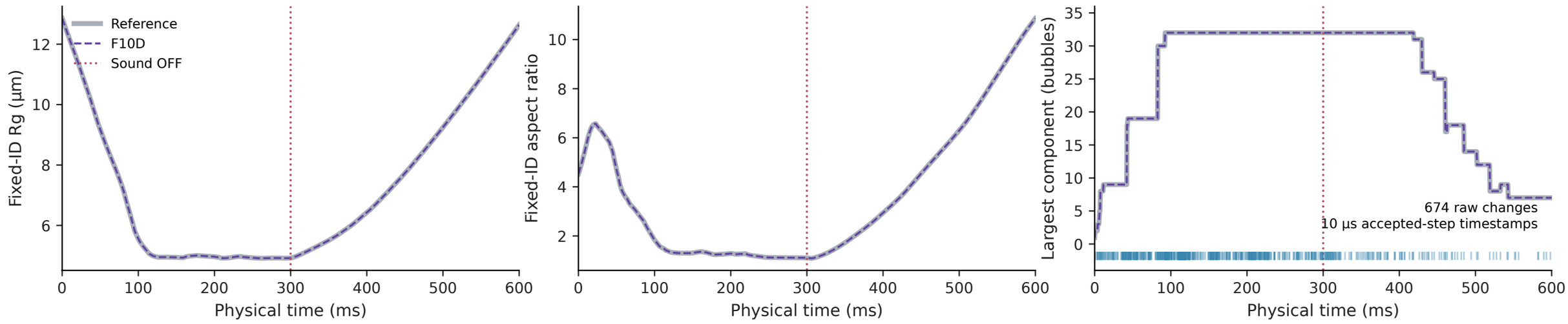
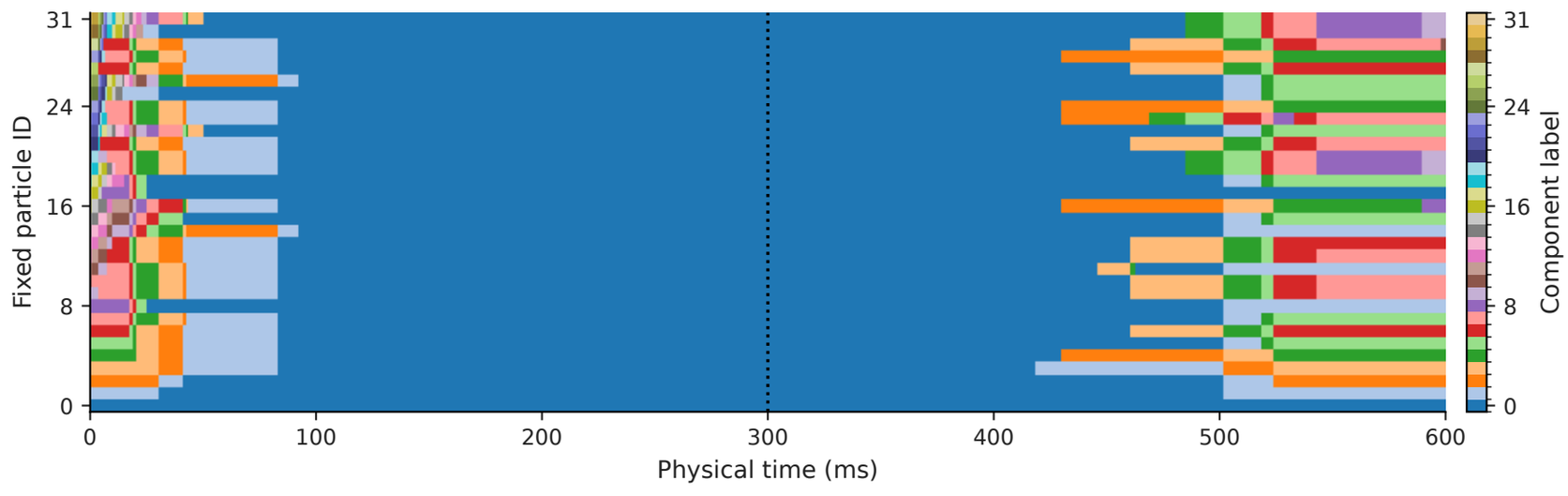
Independent initial arrangements: XY projection, physical radius 1.78 μm, common placement origin

Origin (mm): (-1.979, 0.001, 0.000); color encodes z from -12 to +12 μm (purple to yellow).







i Regression evidence

F5: 1,201 frames over 600 ms
Saved frames every 0.5 ms
32 -> 7 bubbles in largest group
(after sound OFF at 300 ms)

33 persistent merges
11 persistent splits; 1 right-censored
All labels / masks / events exact
Same-server Rg / tensor / Kabsch exact
Max axis-angle error: 2.71e-07 deg
Preregistered tolerance: 1e-5 deg

Top row: actual generated states before preparation, no interpolation or geometric rescaling. F5 overlays use identical saved input.
Colors in panel h denote frame-local component labels, not persistent cluster identities. Graph split/merge does not mean bubble rupture/fusion.
EXPLORATORY / PHYSICS_UNCALIBRATED; independent-seed failures retained; training_ready = false.